DE–7449 **13**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

DIGITAL COMPUTER FUNDAMENTALS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. (a) Explain the conversion process from binary number system to other number system.

   (b) Explain various binary codes with an example.

2. (a) Explain de Morgan's theorems with an example.

   (b) Discuss encoder with neat diagram.

3. Simplify the Boolean function :

   $F(A, B, C, D, E) = (0,2,4,9,11,13,15,17,21,25,27,29,31)$.

4. Design a BCD-to-access-3 code converter.

5. Discuss the full adder with an example.

6. Write a logic diagram of clocked R-S flip-flop with an example.

7. Show the logic diagram of clocked D-flip-flop with AND and NOR gates.

8. Write short note input-output units.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Discuss the following :

   (a) Convert 237.245 octal number into other number systems.

   (b) 1's and 2's complement

10. (a) Obtain the simplified expression in sum of products for the following Boolean function :

   $$A'B'C'D' + AC'D' + B'CD' + A'BCD + BC'D$$

   (b) Discuss demultiplexer with an example.

11. Design a circuit that compares two 4-bit numbers, A and B, to check if they are equal. The circuit has one output X, so that $X = 1$, A = B and $X = 0$ if A not equal to B.

12. Explain D-flip-flop and draw the logic of a master-slave D-flip-flop.

13. Explain shift register.

14. Explain BCD counter.

15. Write short notes on :

   (a) Computer memory

   (b) Processor.

———————

DE–7450 **14**

DISTANCE EDUCATION

M.Sc. (5 YEARS INTEGRATED) DEGREE EXAMINATION, DECEMBER 2013.

Software Engineering

C AND DATA STRUCTURES

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Give the character set of C. Mention the important keywords in C.

2. What are the loop statements used in C? Write a program using for statement.

3. Discuss the definition of functions, with example.

4. Define macros with examples.

5. Explain array with an example program.

6. How to process a data file?

7. Discuss stack and its applications.

8. Explain linked list and its operations.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. (a) Explain data types with an example.

   (b) Explain library functions.

10. (a) Write a C program to generate Fibonacci series.

    (b) Write a C program to find the factorial value.

11. Write a C program to arrange the given numbers in ascending order.

12. Explain pointers with an example.

13. Discuss dynamic memory allocation.

14. Discuss queue operations with an example.

15. Write short notes on:

    (a) Singly linked list

    (b) Binary tree

    (c) Binary tree applications.

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DE–7451 **15**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2013.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Construct the truth table of the following formulas.

   (a) $\neg(P \vee (Q \wedge R)) \rightleftharpoons ((P \vee Q) \wedge (P \vee R))$

   (b) $P \wedge (P \vee Q)$.

2. Show that the set inclusion is a partial ordered relation.

3. Give the power set of the following sets

   (a) $\{a, \{b\}\}$

   (b) $\{1, \emptyset\}$

   (c) $\{a, b, c\}$.

4. Let $f : A \to A$ be a function, where $A = \{1, 2, 3, ..., n\}$. Prove that f is $1-1$ if and only if f is onto.

5. Let $f : X \to Y$ be a function and $A, B \subset X$. Show that $f(A \cup B) = f(A) \cup f(B)$ and $f(A \cap B) \subset f(A) \cap f(B)$.

6. (a) Give an example of a semigroup but not a moniod.

   (b) Give an example of a moniod but not a group.

   (c) Give an example of a group but not an abelian group.

7. Show that every cyclic group of order n is isomorphic to the group $\langle \mathbf{Z}_{n,} +_n \rangle$.

8. Define graph and digraph. Give an example to each.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write an equivalent formula for $P \wedge (Q \rightleftharpoons R) \vee (R \rightleftharpoons P)$ which does not contain the conditional and biconditional.

10. Let R and S be equivalence relations of X. Does $R \cap S$ and $R \cup S$ equivalence relations? Justify your answer.

11. Let $R = \{\langle 1,2\rangle, \langle 3, 4\rangle, \langle 2, 2\rangle\}$ and $S = \{\langle 4, 2\rangle, \langle 2, 5\rangle, \langle 3, 1\rangle, \langle 1, 3\rangle\}$. Find $R \circ S$, $S \circ R$, $R \circ (S \circ R)$, $(R \circ S) \circ R$, $R \circ R$, $S \circ S$ and $R \circ R \circ R$.

12. If $f : X \to Y$ and $g : Y \to Z$ are both 1-1 and onto, then prove that $g \circ f$ is both 1-1 and onto and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

13. (a) State and Prove Lagrange's theorem.

    (b) If G is a finite group of order n, then prove that $a^n = e$, for all $a \in G$.

14. Let $\langle M, * \rangle$ be a monoid. Then prove that there exists a subset $T \subset M^M$ such that $\langle M, * \rangle$ is isomorphic to the monoid $\langle T, \circ \rangle$.

15. (a) Define strong component for a simple diagraph.

    (b) Prove that every node of the simple digraph lies in exactly one strong component.

DE–7452 **23**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

OBJECT ORIENTED PROGRAMMING AND C++

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write short notes on Object Oriented languages.

2. Discuss briefly about basic and user defined data types in C++ with examples.

3. Explain inline functions with an example.

4. Write a C++ program to illustrate the use of a static data members.

5. Explain the concept of Pointer to objects.

6. How is polymorphism achieved at compile time and runtime?

7. Briefly explain the Hierarchical inheritance with example.

8. What is a Virtual base class? When do we make a class virtual?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the basic concepts of Object Oriented Programming.

10. Discuss various operators in C++.

11. Describe Multiple Constructors in a class through an example.

12. (a) Discuss about dynamic constructor.

    (b) Write a C++ program to illustrate how pointers to a derived object are used.

13. (a) Explain the overloading unary minus with an example.

    (b) Explain function overloading with suitable example.

14. (a) Write a C++ program to illustrate hybrid inheritance.

    (b) What are virtual functions? Explain them with an example C++ program.

15. Discuss on the following :

    (a) Derived and base classes.

    (b) Static and dynamic binding.

DE–7453 **24**

DISTANCE EDUCATION

M.Sc. (SE)(5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

OPERATING SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Which type of services are provided by OS in multiple environment?

2. Explain some system calls in an operating system.

3. With the help of an example, explain how race conditions can be avoided?

4. What is process management? Discuss.

5. Write short notes on Interrupt handlers.

6. Write notes on memory management with bit-maps.

7. What is swapping? Discuss.

8. What is a file? Discuss the operation on files.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Explain real time operating system. Compare it with other operating systems.

10. What is semaphore? Discuss any two applications of semaphore.

11. Explain round Robin and first come first served scheduling algorithm.

12. Explain the Banker's algorithm to avoid deadlock giving an example.

13. Write a short notes on the following :

    (a) Buddy system

    (b) Paging

14. (a) Discuss about multiprogramming with fixed partitions.

    (b) Explain the address translation mechanism in paging.

15. Discuss on the following :

    (a) Single-level directory

    (b) Protection mechanism

DE–7454 **25**

DISTANCE EDUCATION

M.Sc.(Software Engineering) (5 Years Integrated)DEGREE EXAMINATION, DECEMBER 2013.

RELATIONAL DATABASE MANAGEMENT SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. Define schema and subschema. Give examples.
2. Write short notes on mapping operation. Give examples.
3. Discuss about data control facilities.
4. What are the various types of databases? Explain.
5. What are the benefits of Oracle? Explain
6. Discuss database objects. Give example.
7. How to create a table? Give example.
8. Define Indexes. Give example.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. What are the advantages and disadvantages of DBMS? Explain.
10. Explain about
    (a) DML facilities
    (b) DDL facilities.

11. Write the overview of Oracle architecture.

12. What are the various set operations in Oracle? Give example.

13. Explain about various SQL commands.

14. Explain about embedded SQL.

15. (a) How to join multiple tables in a query? Give examples. (8)

    (b) Discuss about subqueries in Oracle. Give examples. (7)

DE–7455 **31**

DISTANCE EDUCATION

M.Sc. (Software Engineering)(5 year Integrated) DEGREE EXAMINATION, DECEMBER 2013.

COMPUTER NETWORKS

Time : Three hours Maximum : 100 marks

SECTOIN A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain about broadband and narrow band ISDN.

2. Discuss about TCP/IP reference model with a neat diagram.

3. Write short notes on ALOHA.

4. Discuss about error correcting codes with an example.

5. Discuss about the service categories of network layer.

6. Explain the network layer design issues in detail.

7. Write a note on TCP.

8. Discuss about WWW in detail.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about transmission media.

10. Discuss about the telephone system with neat diagrams.

11. Discuss about sliding window protocols.

12. Explain about media access sublayer in detail.

13. Discuss about congestion control algorithms in detail.

14. Discuss about UDP protocol.

15. Explain about E–mail.

DE–7456 **32**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

INTERNET AND JAVA PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Describe Telnet concept in detail.

2. How is an interface defined? Give an example.

3. How does try … catch mechanism handle an exception?

4. Write applets to draw the following shapes

   (a) Square inside a circle

   (b) Circle inside a square.

5. List the various event handlers available for handling the 'Mouse Event'.

6. List any five form elements in detail.

7. What are the salient features of RMI techniques?

8. What is socket? How is it used in networking?

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Define internet. What are the components required for making a new internet connection?

10. Explain Oops concepts in detail.

11. How to implement multiple inheritance in java?

12. Define exception. How to create user defined exception?

13. List out any five AWT controls in detail.

14. What is a socket? How is it used in networking?

15. (a) Define array. How to create arrays in java script?

    (b) Explain control structures in java script.

DE–7457 **33**

DISTANCE EDUCATION

M.Sc. (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

SOFTWARE ENGINEERING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Illustrate spiral model in detail.

2. Describe software development process model.

3. Explain project estimation concept in detail.

4. Explain cost factor of software development in detail.

5. Explain project scheduling plan in detail.

6. Explain software reviews concept in detail.

7. Describe Information flow concept in detail.

8. Explain Interface Design.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Explain

   (a) RAD

   (b) s/w process.

10. Describe process metrics concept in detail.

11. Describe software cost estimation concept in detail.

12. What are all the resources required for the s/w development?

13. Explain software configuration management in detail.

14. Describe prototyping models in detail.

15. Explain :

   (a) White box testing

   (b) Control Structure Testing

   (c) Block box testing.

DE–7458 **34**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

COMPUTER GRAPHICS AND MULTIMEDIA

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain output primitives.

2. Explain circle drawing algorithms in detail.

3. Discuss about polygons, splines and bezier curves.

4. Explain various color models.

5. Write in detail about multimedia hardware and software.

6. Discuss about various audio and video formals.

7. Explain multimedia communication systems.

8. Discuss about multimedia database management systems.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the I/O devices along with internal processing.

10. Discuss about DDA and line drawing techniques.

11. Explain various 2D transformations and composite transformations.

12. Discuss about projections and clippings.

13. Discuss synchronization issues in multimedia.

14. Explain and multimedia project development.

15. Explain the following :

    (a) Videos conferencing

    (b) Virtual Reality

    (c) Media on demand.

DE–7459 **35**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

UNIX AND SHELL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the following Unix commands.

   (a) man

   (b) uname

   (c) who

   (d) is.

2. Briefly discuss about text processing utilities in Unix.

3. Write an example for PIPE and explain.

4. How do you perform arithmetic in shell script?

5. Explain how to work with define functions.

6. Discuss about arrays in Unix.

7. Explain korn shell features and files.

8. Explain the file concepts in C-shell programming.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Write about Unix utilities functions.

10. Briefly explain Unix architecture, features and security.

11. Discuss about various filters in Unix.

12. Discuss about the following :
    (a) String functions
    (b) Mathematical functions.

13. Discuss about korn shell.
    (a) Basic scripts format
    (b) Decision making controls.

14. Write about scripts in C-shell programming.

15. Explain directory API in C-shell.

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DE–7663 **16**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

LAB I — C PROGRAMMING LAB

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by lot system.

1. (a) Write a C program to solve quadratic equation.

   (b) Write a C program to convert a sentence in to proper sentence. Example : Input. "This is a wonder land". Output : "This Is A Wonder Land".

---------- Cut here ----------

2. (a) Write a C program to find all possible roots of a quadratic equation using function definition.

   (b) Write a C program to print the integral value and fractional value in a number. Example : For 29.84, the output is integral value : 29 & Fractional value : 84.

---------- Cut here ----------

3. (a) Write a C program to reverse a string without using string functions.

   (b) Write a C program to Add and Subtract two matrices.

4. (a) Write a C program to find the sign of a number.
   (b) Write a C program to multiply two matrices.

---------- Cut here ----------

5. (a) Write a C program to print the ascending order of array elements.
   (b) Write a C program to print the result of the examination for 5 students using structure. The result is to be computed as "Pass", for mark greater than or equal to 35, otherwise "fail".

---------- Cut here ----------

6. (a) Write a C program to show the use of any five C mathematical functions.
   (b) Consider the following. "A visiting professor paid as Rs. 1,000/- for conducting 4 hours of lecture class per day." Thereafter, overtime paid as Rs. 300/- per hour, subject to maximum of Rs. 600/- per day. The week days are from Monday to Friday. Write a C program to print the weekly report as detailed below : (40)

Week Days	No. of contact classes	Over-time hours	Claim of the day
1) Name of the Professor : _____ & Dept. _____.			
Sun.-to-Sat.	---	---	---
		Week Total : Rs.	
2) Name of the Professor : _____ & Dept. _____.			
Sun.-to-Sat	---		---
		Week Total : Rs.	
3)	...		
		Week Total : Rs.	
		Over all Total : Rs.	

7. (a) Write a C program to read the elements of an array. Use pointer to locate and display each value.

   (b) Write a C program to find a word as palindrome or not.

---------- Cut here ----------

8. (a) Write a C program to find identity matrix.

   (b) Write a C program to implement all string operations using switch statement.

DE–7664 **17**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

DATA STRUCTURES LAB

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by lot system.

1. (a) Write a C program to insert and delete elements in a queue using array.

   (b) Write a C program to evaluate mathematical expression using stack.

---------- Cut here ----------

2. (a) Write a C program to sort one dozen number in ascending order and show variable and the values before and after sorting.

   (b) Write a C program to insert and delete elements from stack.

3. (a) Write a C program to sort an array using insertion sort.

(b) Write a C program to evaluate mathematical expression using stack.

---------- Cut here ----------

4. (a) Write a menu driven program in C to find an element using Binary search method.

(b) Write a C program to copy contents of one file to another.

---------- Cut here ----------

5. (a) Write a C program to insert and delete elements in a linked list.

(b) Write a C program to insert and delete elements in a queue using array.

---------- Cut here ----------

6. (a) Write a C program to do push and pop operations.

(b) Write a C program to sort an array using selection sort.

7. (a) Write a C program to sort an array using selection sort.

(b) Write a C program for linked list implementation of Queue.

---------- Cut here ----------

8. (a) Write a C program to sort half dozen numbers in descending order and show the values before and after sorting.

(b) Write a menu driven program in C to find an element using Linear search method.

DE–7665 **26**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2013.

C++ LABORATORY

Time : Three hours Maximum : 100 marks

Examine has to choose any one question.

Break up of marks

Record	: 10
Algorithm	: 10
Program	: 50
Debugging	: 10
Execution	: 10
Result	: 10
Total	: 100

1. Write a C++ program to find the Armstrong numbers between 100 and 999.

---------- Cut here ----------

2. Write a C++ program to produce the following output.

```
1
2  2
3  3  3
4  4  4  4
5  5  5  5  5
```

3. Write a C++ program to add 2 matrices of order 3×3 using pointer method.

---------- Cut here ----------

4. Write a C++ program to find the factorial of given 2 numbers using single function call.

---------- Cut here ----------

5. Write a C++ program to implement the concept of constructor with an example.

---------- Cut here ----------

6. Write a C++ program to implement the concept of multiple inheritance with an example.

---------- Cut here ----------

7. Write a C++ program for implementing function overloading with an example.

---------- Cut here ----------

8. Write a C++ program to determinant of a given matrix.

---------- Cut here ----------

9. Write a C++ program to calculate string length using pointers.

---------- Cut here ----------

10. Write a C++ program to create a class called student that contains roll number, name, mark1, mark2, mark3. Include a member function to get data from user of 'n' students. The output should be in the following format : Student roll no name M1 M2 M3 total average result grade.

DE–7666 **27**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated)
DEGREE EXAMINATION, DECEMBER 2013.

RELATIONAL DATABASE MANAGEMENT SYSTEM LAB

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

1. (a) Create a table STUDENT contains :

Roll no, name, mark1, mark2, mark 3, Using the above table, do the following:

(i) List out the students whose name starts with 'S'.

(ii) List out the student's detail who got highest total.

(iii) For passing minimum, the student should get atleast 50 marks in all subjects. Set the grade for the students as follows:

* 'DISTINCTION' if total > 275

* 'A' if total between 250 and 275

* 'B' if total between 200 and 250

* 'C' if total between 150 and 200.

Display all the students who have got 'DISTINCTION' and grade 'A'.

(b) Create a table EXPERIENCE with the following fields :

College-name, professor-code, professor-name, qualification, experience, sex and address.

Write Queries to do the following:

(i) List out the professors who have more than 15 years experience and have completed Ph.d.

(ii) List out the female professors who have completed Ph.d and having experience greater than or equal to 10 years.

(iii) Display all those records whose names ends with "a" and experience greater than or equal to 5 years.

---------- Cut here ----------

2. (a) A railway reservation counter contains the following database.

RESERVATION database with fields:

Train-no, class, passenger-name, seat-no, Birth, Sex, age and marital-status.

Write Queries to do the following:

(i) List the passenger names with seat number whose age is greater than or equal to 60.

(ii) List the total number of married female passengers in the train.

(iii) Set the passengers as "ADULT" whose age is between 18 to 25.

(b) Create an EMPLOYEE table with (emp-no, name, dept, salary, designation, date-of-birth, date-of-joining).

Write the query for the following.

(i) Find out the average salary of the employees of each department name-wise.

(ii) List out the employee who is getting minimum salary working in "EDP" dept.

(iii) List out the employee who is getting maximum salary working in "PHYSICS" dept.

---------- Cut here ----------

3. (a) Create a table EMPLOYEE with the following fields.

EName, ENO, Basic, HRA, DA, PF, Netpay.

HRA = 20% of Basic pay

DA = 12% of Basic pay

PF = 8% of Basic pay

Netpay = Basic + HRA + DA – PF

Write a PL/SQL program to prepare salary report for 10 employees using cursor.

(b) Write Queries to do the following:

(i) Display the employee's details who gets the highest netpay.

(ii) Display the employees whose names starts with "S" along with PF.

(iii) Sort and display the employee details in descending order of Basic pay.

4. Write a PL/SQL program for Hospital Management System with the following fields:

   Patient-no, Patient-name, Age, Patient-type (IN/OUT), Consulting-Doctor, Charge, Blood-test-charge, Xray-Charge and total-fee.

   Prepare reports with the following conditions:

   (i) The patients under OUT-PATIENT category.

   (ii) The patients with age 60 and above.

   (iii) Find the overall total fees.

---------- Cut here ----------

5. An egg vendor buys eggs on wholesale from a Distributor for Rs.2 and sell them in retail for Rs.2.50. At the end of the day, the damaged eggs are returned to the distributor for a Re 1 rebate per egg. Write a program using oracle to prepare a report for the egg vendor in the following format for 1 week.

Day	Bought	Sold	Return	Profit	Loss
1	xxx	xx	xx	xxxx	xxxx
2	xxx	xx	xx	xxxx	xxxx
3	xxx	xx	xx	xxxx	xxxx
4	xxx	xx	xx	xxxx	xxxx
5	xxx	xx	xx	xxxx	xxxx
6	xxx	xx	xx	xxxx	xxxx
7	xxx	xx	xx	xxxx	xxxx
Total					

DE–7667 **36**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

INTERNET AND JAVA LAB

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Each subdivision carries 50 marks.

1. (a) Write a Java script to create a Order Form to select the house articles.

   (b) Write a Java program using applet to display any 3 images when 3 buttons in the border layout are clicked. The image should be displayed at the centre.

---------- Cut here ----------

2. (a) Write a Java program using Applet to create the frames and its control.

   (b) Write a Swing program to create the Tabbed panels.

---------- Cut here ----------

3. (a) Write a Menu-Drive java program to do database functions using database connectivity facility. The functions should display, edit, insert and delete a record from a table.

   (b) Write a Java program to create a color as the background.

4. (a) Write a Java script to create a color palet and display the text in the color chosen from the palet with proper background color.

(b) Develop a picture gallery having atleast 5 pages. Each of them is having several pictures with suitable information.

---------- Cut here ----------

5. (a) Write a Java program using Applet to display the different color and fonts.

(b) Write a Swing program to create Buttons with:

(i) Tool tip text

(ii) Image

(iii) Border

(iv) Short cut key.

DE–7668 **37**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2013.

UNIX AND SHELL PROGRAMMING LAB

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Each subdivision carries 50 marks.

1. (a) Write a Menu driven Shell program for the following :
   (i) List of files
   (ii) Processes of users
   (iii) Today's date
   (iv) Users of system
   (v) Quit to UNIX.

   (b) Write and execute a shell program to send mail to groups of users by extracting their Ids from/etc/group file.

---------- Cut here ----------

2. (a) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :
   (i) File existence
   (ii) File readable
   (iii) File writable
   (iv) Both readable and writable

   (b) Write and execute a shell program to generate leave status report of an organization.

3. (a) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display "Pattern found" else display "Error message". Also cheek if right number of arguments are entered.

(b) Write a shell program to sum up the series :

$$\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{10!}$$

---------- Cut here ----------

4. (a) Write a shell program which periodically monitors the disk for the existence of a file and then executes the program once the file has been located (using UNTIL statement).

(b) Write a program for four-function calculator by using yacc command.

---------- Cut here ----------

5. (a) Write a shell program which accepts the name of a file from the standard input and then perform the following tests on it :

(i) Enter the 5 names in a file

(ii) Sort the names in existing file

(iii) List unsorted and sorted file

(iv) Quit

(b) Write and execute a shell program to print monthly payslips for employees in an organization. Assume your own data and generate accordingly.